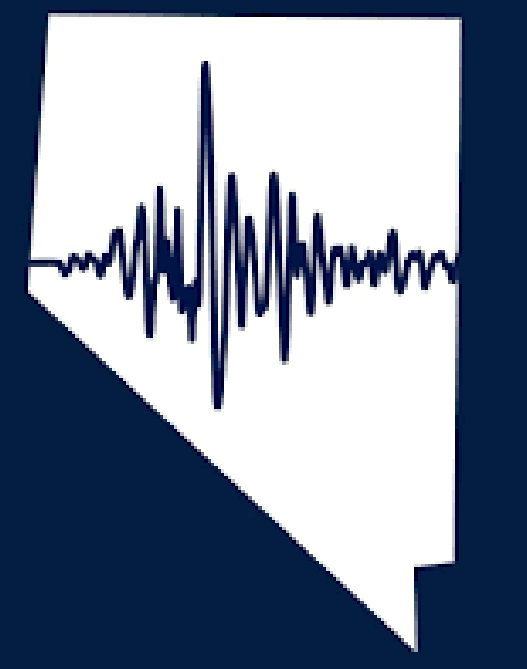




USING DYNAMIC RUPTURE SIMULATIONS TO IMPROVE SEISMIC HAZARD MODELS FOR NORMAL-SLIP EARTHQUAKES

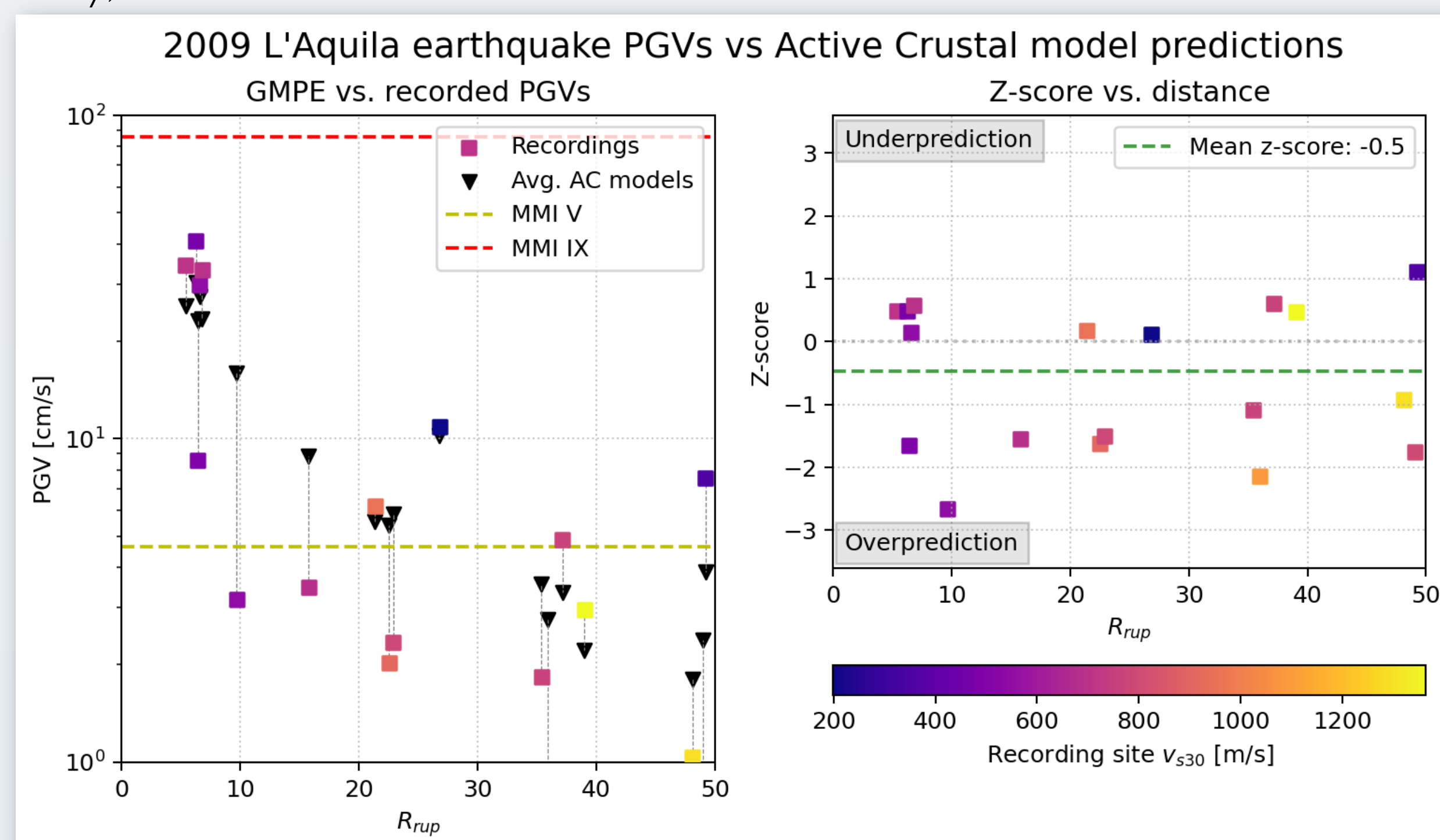


YAIR FRANCO, DANIEL TRUGMAN
NEVADA SEISMOLOGICAL LABORATORY, UNIVERSITY OF NEVADA, RENO

BACKGROUND

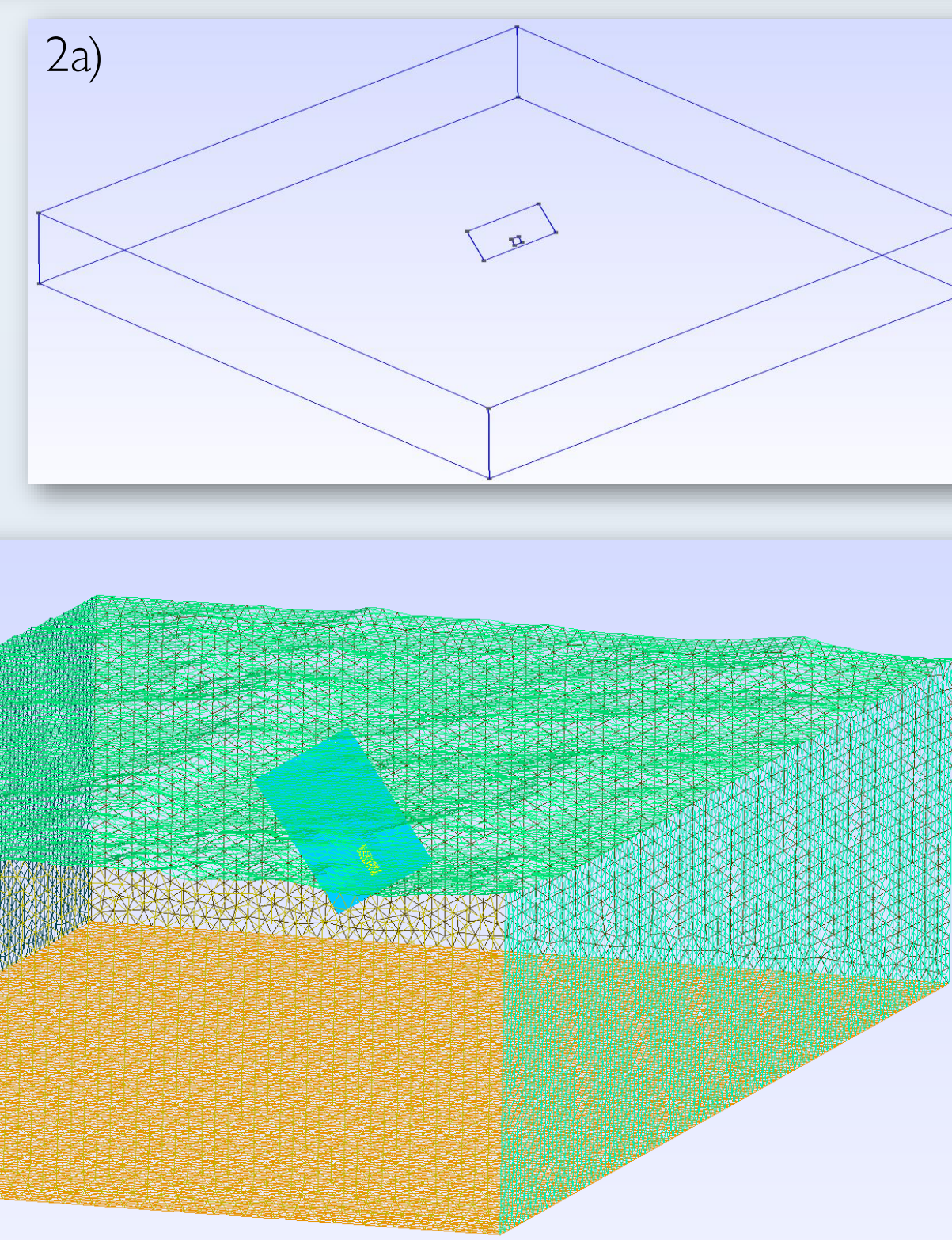
- Normal faults under populated areas are known to have high damage potential
 - 2009 L'Aquila, 2016 Amatrice and Norcia earthquakes
- Normal-slip earthquakes common in U.S. Basin and Range region
 - Scarce data due to sparse instrumentation at the time of largest normal-slip events (1915, 1954)
- 2023 U.S. National Seismic Hazard Model (NSHM) active crustal (AC) ground motion models (GMMs) derived from scarce global records of large normal-slip events
 - ASK 14 dip-slip hanging wall term generalized from kinematic simulations of reverse-slip ruptures
- Data scarcity may lead to hazard mischaracterizations for populations on normal faults, including cities like Reno, Salt Lake City, etc.

Figure 1: Comparison of recorded PGVs and GMM predictions for the M6.3 2009 L'Aquila earthquake. Left plot shows GMDB vs. predicted PGVs vs. R_{rup} (distance from rupture plane). Right plot shows z-score per station.



METHOD

- Analysis of dip-slip rupture dynamics in SeisSol.
- Suite of simulations testing variables:
 - Fault dimensions: 30x15 km
 - Dip: base case 60°
 - Nucleation: 3x3 km high-stress patch
 - Hypocenter: 12 km down-dip, centered along strike
 - Surface rupturing fault
 - Depth-dependent stress parameters
 - No topography, velocity model
- Simulation for large event in Reno area
 - Fault dimensions: 30x15 km
 - Dip: 55°
 - Nucleation: 3x3 km low-friction patch
 - Hypocenter: 9 km depth
 - Surface rupturing fault
 - Depth-dependent stress parameters
 - 3D velocity model and topography



Figures 2a and 2b: Examples of 3D models used for (a) dip-slip experiment suite and (b) Reno scenario with topography.

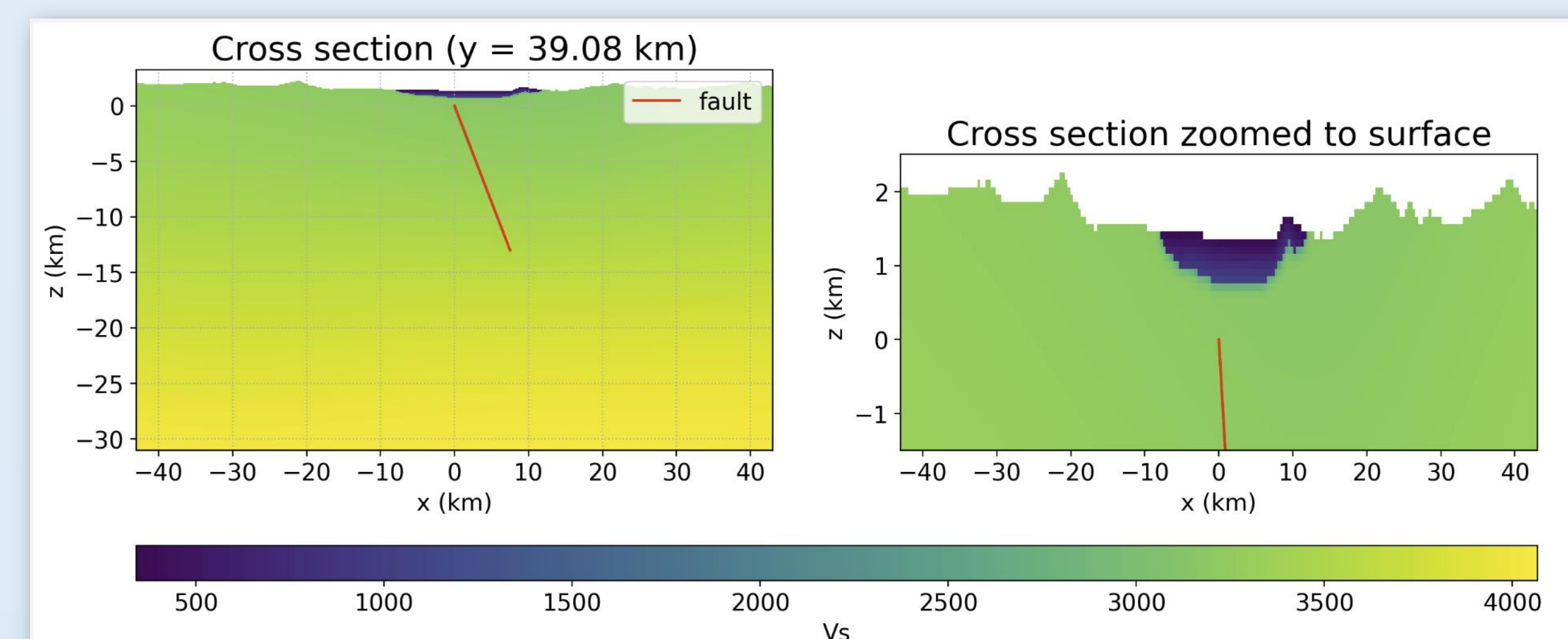
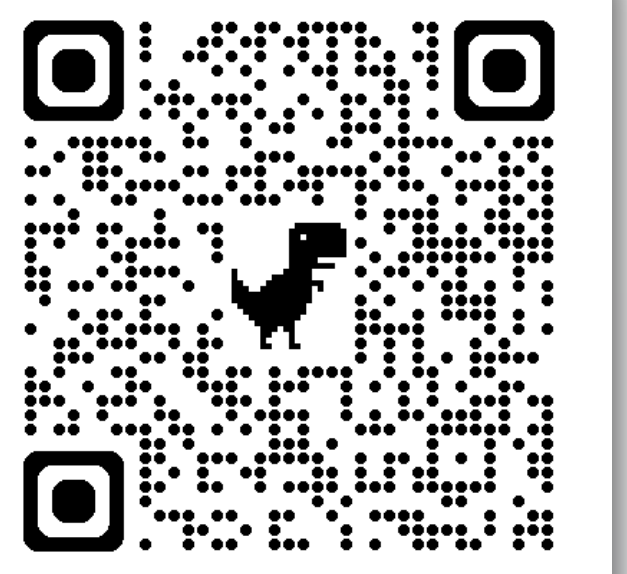


Figure 3: Preview of velocity model used for Reno scenario. This cross section shows the velocities in the basin directly under the Reno area in topography. Elevation is from sea level.

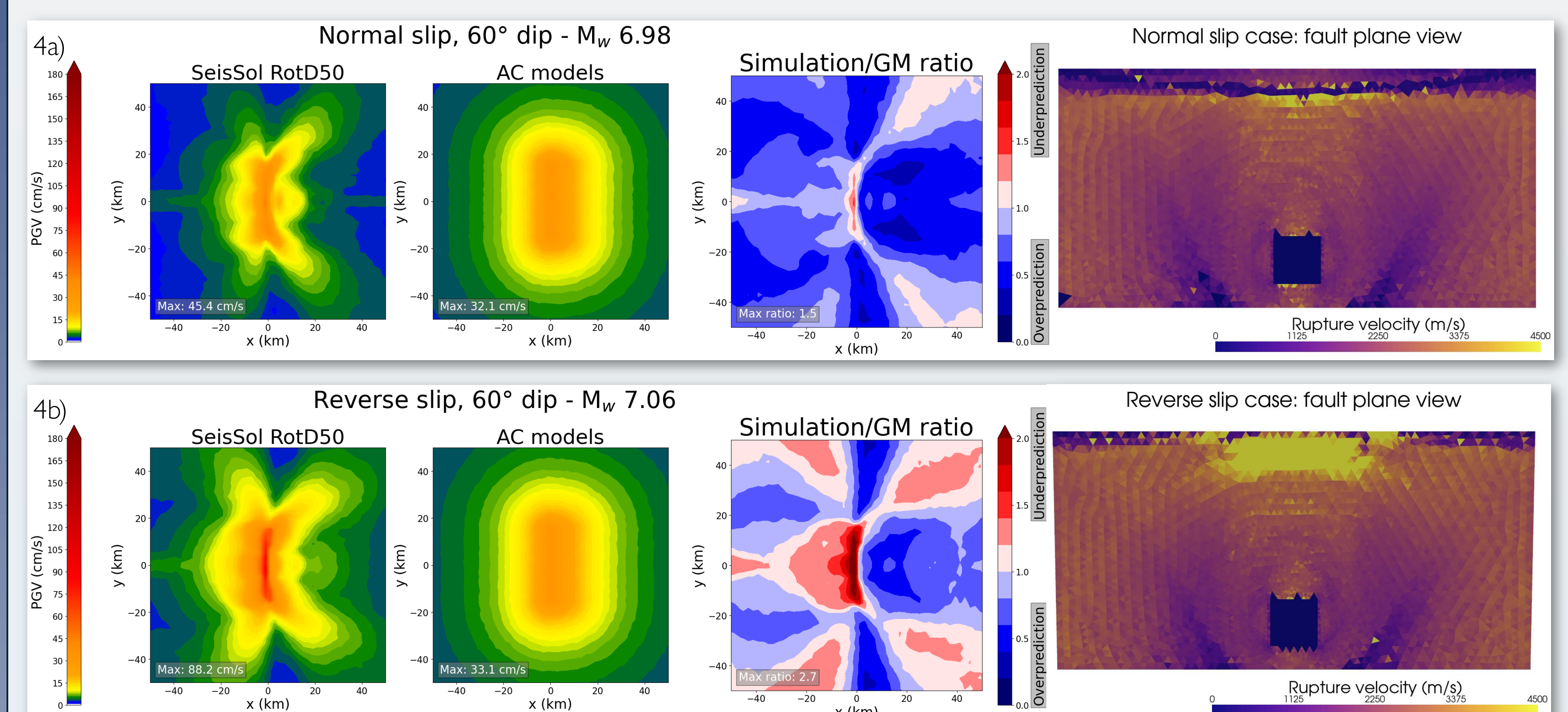
RESULTS

Scan this QR code to view [additional material](https://www.yairfranco.com/scec2026) for this poster at [yairfranco.com/scec2026](https://www.yairfranco.com/scec2026)



Part I: Dip-slip experiment suite

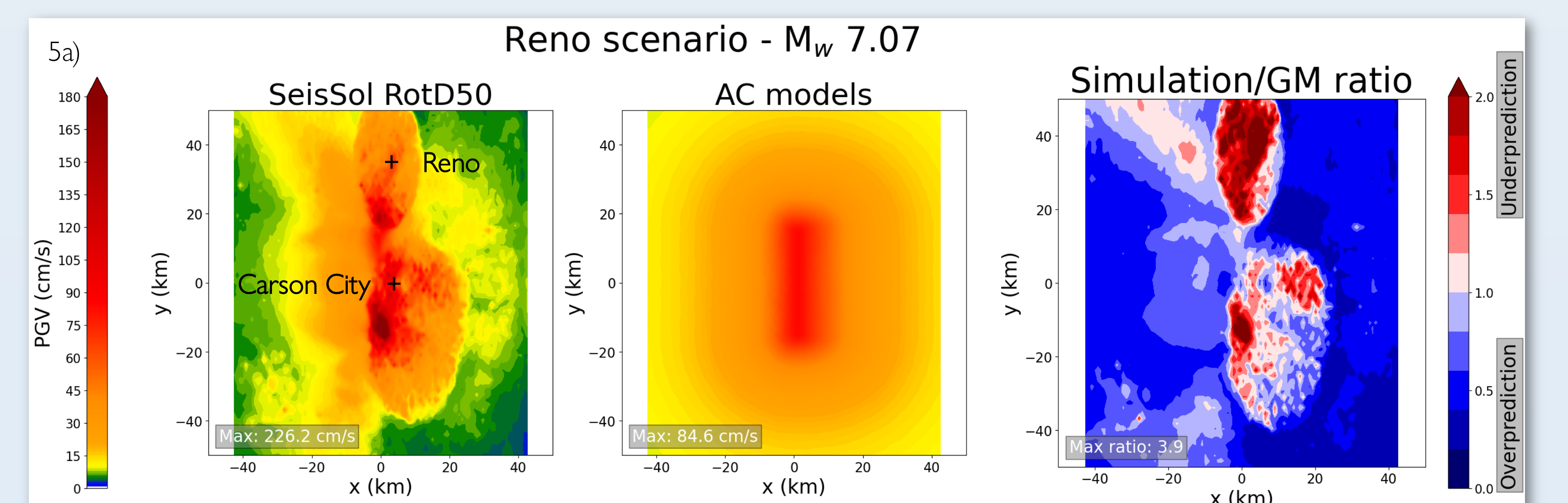
- Near-surface rupture bifurcation phenomenon in normal slip
- Nucleation depth, varying material properties affect rupture propagation
- More intense near-fault ground motion in reverse-slip earthquakes
- Dip and mechanism affect ground motion distribution (see *additional material*)



Figures 4a and 4b: Selected results from the dip-slip experiment suite, with (a) ground motion and rupture velocity for the base, normal-slip, 60° dip case, and (b) the same for an identical reverse slip case.

Part II: Reno scenario (preliminary)

- Several simulation setup issues to improve:
 - Adjust stress field and frictional parameters for lower slip
 - Refine mesh resolution to prevent extreme ground velocities
 - Investigate factors affecting behavior of rupture bifurcation
- Goal: demonstrate basin effects in close proximity to ruptures combined with influence of dynamic rupture phenomena



Figures 5a, 5b, 5c: Results from a preliminary simulation on a M_w 7.3 normal-slip event in the Reno area with (a) simulated vs. GMM ground motions and their ratio, (b) moment release over time, and (c) rupture velocities along the fault plane.